MILITARY SPECIFICLATION

AMPLIFIER, OPERATIONAL
TRIPLE PROGRAMMABLE MICROPOWER

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a monolithic, triple programmable, micropower, operational amplifier.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-M-38510	-	General Specification for Microcircuits
MIL-M-38510/101	-	Microcircuit, Linear, Operational Amplifiers, Monolithic Silicon
MIL-M-55565	-	Microcircuits, Preparation for Delivery of

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:28Z
Check the source to verify that this is the current version before use.

MIL-STD-130	-	Identification marking of U.S. Military Property
MIL-STD-883	-	Test Methods and procedures for Microelectronics

2.1.2 Other Government documents, drawings, and publications. The following other Government documentsform a part of this specification to the extent herein.

DRAWINGS

U. S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9317646 - Qualification Electronic Components

(Copies of specifications, standards, drawings, and publications required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Design and construction.

3.1.1 Outline and dimensions. The outline and dimensions of the finished component shall conform with MIL-M-38510, Case Outline D-2, configuration 1 or 3.

3.1.2 Case. The finished component shall be supplied in a 16 pin, dual-in-line hermetically sealed case. The case shall be of a glass frit seal type, (figure 3) utilizing a ceramic lid, a ceramic base with gold deposit for mounting the die, Alloy 42 or Kovar solid metallic lead frame through to cavity with tin or solder plated external leads, per MIL-M-38510, paragraph 3.5.6.2 A or B, and aluminum cladding and glass frit to accomplish the seal between the Lid, Lead Frame and the Base. The glass frit sealing temperature shall be in excess of 400°C.

3.1.3 Terminal (pin) locations. See figure 1 for pin numbers. Pin 1 location shall be identified by a notch, paint dot or other symbol on the case top.

3.1.4 Solderability. External leads must be acid tin electroplated or solder dipped per MIL-M-38510, paragraph 3.5.6.2 (a) or (b) and if acid tin electroplated, the minimum plating must be 150 microinches. Parts must also be solderable per MIL-STD-883, Method 2003.

3.1.5 Internal lead wire clearance. There shall be no internal lead wires that appear to cross or touch each other when viewed from the vertical axis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:28Z
Check the source to verify that this is the current version before use.

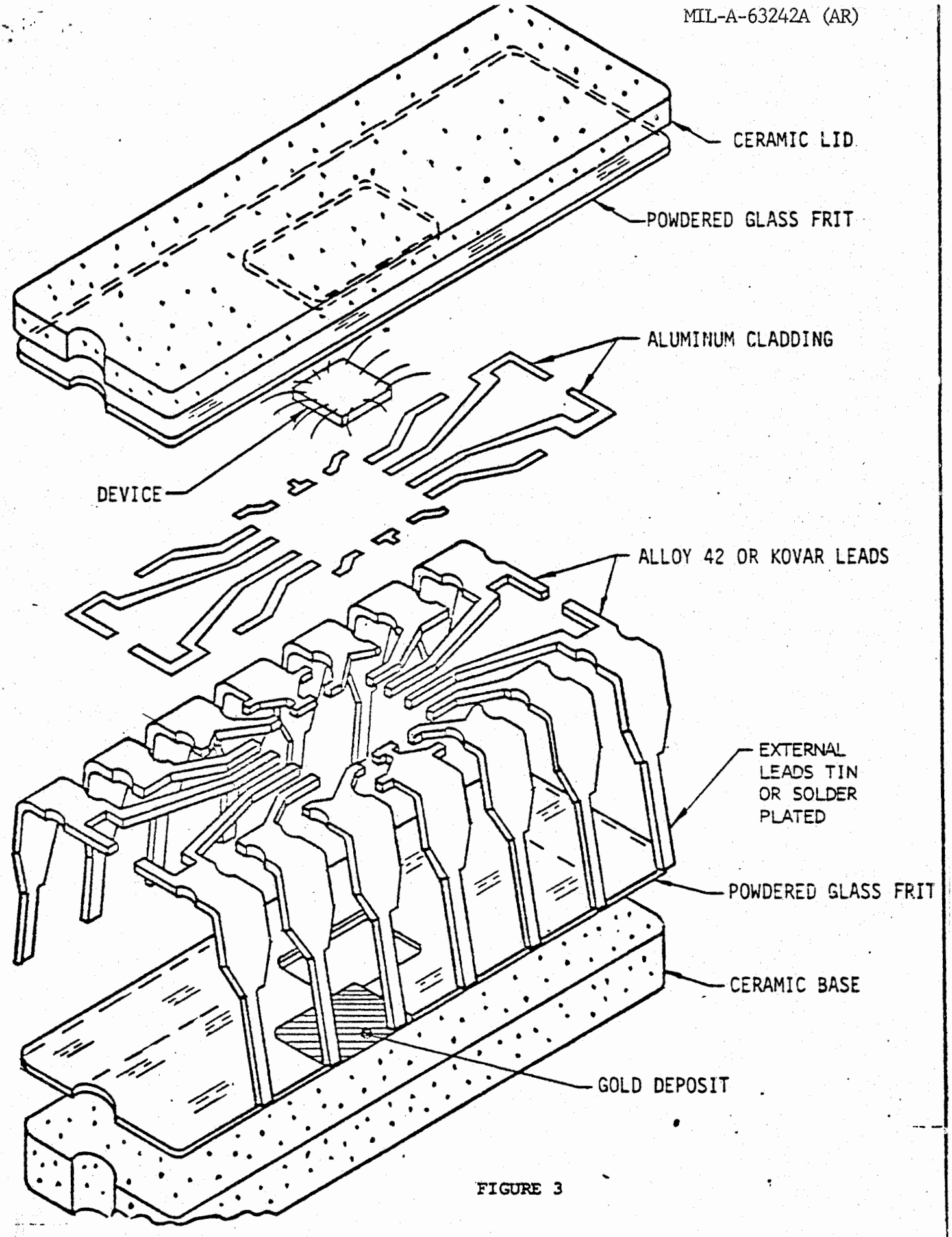


FIGURE 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:28Z
Check the source to verify that this is the current version before use.

3.1.7 Bond strength. Wires from the circuit die to the package header shall be aluminum, .00125 to .0015 inch nominal diameter. The wire bonds shall be capable of meeting the bond strength requirements of MIL-STD-883, Method 2011, Test Conditions D. Pull all bonds on each device tested.

3.1.8 Atmosphere control, storage and handling. The manufacture shall establish and document a control process in which all parts manufactured according to this specification must be scaled and stored in a day atmosphere to preclude moisture entrapment within the I.C. device which could cause subsequent device function degardation or failure. This document shall be submitted 30 days prior to the First Article Inspection to ARRADCOM.

3.2 Temperature.

3.2.1 Operating temperature. The operational amplifier shall comply with all the electrical requirements of this specification over the temperature range of -40°C thru +60°C.

3.2.2 Storage temperature. The operational amplifiers shall comply with all the reqwuirements of this specification after storage over the temperature range of -55°C thru +125°C.

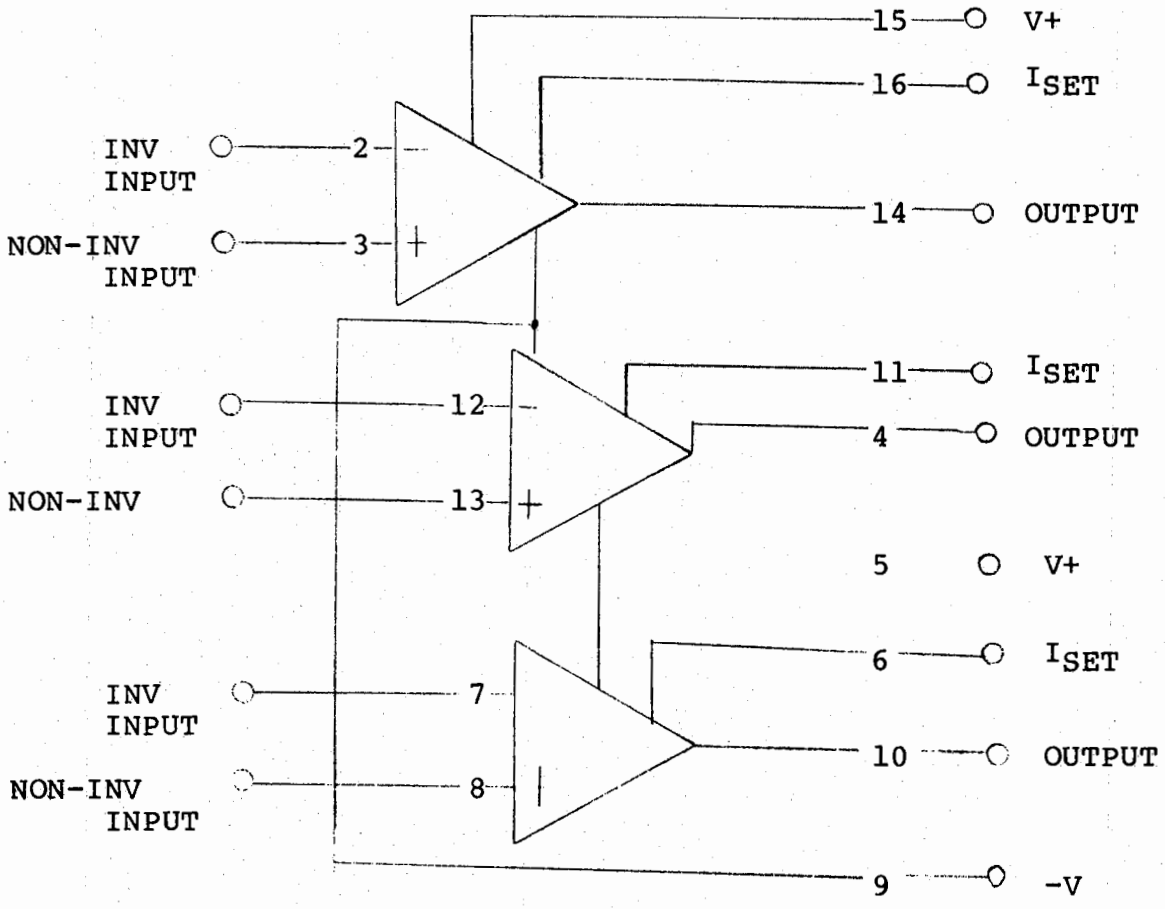


FIGURE 1 - PIN-OUT

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:28Z
Check the source to verify that this is the current version before use.

3.3 Electrical.

3.3.1 Absolute maximum ratings.

Supply Voltage Range	± 7 Vdc
Input Voltage Range	± 7 Vdc 1/
Differential Input Voltage Range	± 7 Vdc 1/
Lead Temperature (Soldering 10 SEC)	± 300°C

3.3.2 Electrical characteristics. Unless otherwise specified, the operational electrical characteristics for each OP AMP shall be as specified in table 1 and the following conditions apply.

V_S = ± 4 volts

I_Q = 30 microamperes

T_A = 25°C except as noted

1/ For supply voltages less than ± 7 Vdc, the absolute maximum input voltage is equal to the supply voltage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:28Z
Check the source to verify that this is the current version before use.

TABLE I

ELECTRICAL CHARACTERISTICS

PARAMETERS	CONDITIONS	LIMITS	UNITS
T_A = 25°C			
INPUT OFFSET VOLTAGE (V_{IO})	$R_S \leq$ 100K	6.0	MV MAX
INPUT OFFSET CURRENT (I_{IO})		10	NA MAX
INPUT BIAS CURRENT (I_{IB})		30	NA MAX
INPUT RESISTANCE		3	M MIN
POWER CONSUMPTION (P_D)	$V_{O=0}$. R_{SET} 1.3M	400	MAX
LARGE SIGNAL VOLTAGE GAIN (A)	R_L = 10K	50	V-MV MIN
OUTPUT VOLTAGE SWING	R_L = 20K OHMS $V_S = \pm 4V$	$\pm$ 3	V MIN
-40°C $\leq T_A \leq$ +60°C			
INPUT OFFSET VOLTAGE	$R_S \leq$ 10K OHMS	7.5	MV MAX
INPUT OFFSET CURRENT		15	NA MAX
INPUT BIAS CURRENT		50	NA MAX
LARGE SIGNAL VOLTAGE GAIN	R_L = 10K OHMS	50	V-MV MIN

3.4 Screening. Prior to final inspection, the finished component shall be screened in accordance with Table II except that internal visual (pre-cap) shall be performed prior to capping. Components subjected to the screening procedures shall be capable of functioning and meeting the requirements of 3.3.

3.5 External visual. The finished component shall be examined for the requirement of MIL-STD-883, Method 2009.

3.6 Qualification and acceptance tests. OP amps described herein shall comply with the qualification requirements described in Table III and the lot by lot Acceptance Tests described in Table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:28Z
Check the source to verify that this is the current version before use.

TABLE II

SCREEN	METHODS 2/	REQUIREMENT
Internal Visual (pre-cap)	2010, Test Condition B	100%
Stabilization Bake	1008, 24 Hrs. Min. Test Cond C Min (See 3.3.1)	100%
Temperature Cycling	1010, Cond. C, 10 Cycle	100%
Centrifuge	2001, Test Cond. E Y_1 axis only	100%
Hermeticity		
A. Fine	1014, Test Cond. A or B	100%
B. Gross	1014, Test Cond. C	100%
Burn-In Test	1015, @ 125°C for 168 hours (See Para 4.3.3.1)	100%
Post Burn-In	1015, @ 25°C All Electrical Parameters per Table I	100%

2/ Methods refer to MIL-STD-883 except as noted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:28Z
Check the source to verify that this is the current version before use.

TABLE III - QUALIFICATION TESTS

SYMBOL	EXAMINATION OR TEST	CONDITION	REQ PARA	METHOD PARA	LIMITS MIN	LIMITS MAX	LIMITS UNITS	NO. OF DEVICES TO BE INSPECT	NO. OF DEFFECT ALLOWED
	GROUP I								
	SCREEN TESTING		4.2.3	TABLE II				100%	0
	MARKING		4.2.9	3.				3	0
	BOND STRENGTH		4.2.5	2011				6 DEVICES ALL BONDS	0
	INTERNAL VISUAL (DELID)			2010				6	0
	PHYSICAL DIMENSIONS		4.2.8	2016				6	0
	SOLDERABILITY		4.2.6	2003				6 DEVICES ALL LEADS	0
	DIE SHEAR STRENGTH		4.2.7	2019				6	0
	THERMAL SHOCK		4.2.11	1011 15 CYCLES				52	4
	GROUP II	(T_A = +25°C)	4.2.4.1	4.3.4 3.3.2				77	1
	ELECTRICAL CHARACTERISTICS TABLE I								
	GROUP III	(-40°C +65°C)	4.2.4.2	4.3.4 3.3.2				77	1
	ELECTRICAL CHARACTERISTICS TABLE I								

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:28Z
Check the source to verify that this is the current version before use.

TABLE IV - ACCEPTANCE TESTS

SYMBOL	EXAMINATION OR TEST	CONDITION	REQ PARA	METHOD PARA	LIMITS MIN	LIMITS MAX	LIMITS UNITS	NO. OF SAMPLES TO BE INSPECT	NO. DEFE ALLO
	GROUP I								
	SCREEN TESTING		4.2.3	TABLE II				100%	0
	MARKING		4.2.9	3.6				3	0
	SOLDERABILITY		4.2.6	2003				5 DEVICES ALL LEADS	0
	PHYSICAL DIMENSIONS		4.2.8	2016				3	0
	BOND STRENGTH		4.2.5	2011				5 DEVICES ALL BONDS	0
	DIE SHEAR STRENGTH		4.2.7	2019				5	0
	THERMAL SHOCK			1011 15 CYCLES				52	4
	EXTERNAL VISUAL			2009				1.0% AQL	
	GROUP II (T_A = +25°C) (& END POINT TESTS) PER TABLE III, GROUP II							77	1
	GROUP III (-40°C, +65°C) PER TABLE III, GROUP III							52	2

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:28Z
Check the source to verify that this is the current version before use.

3.7 Country of manufacture. Operational amplifiers described herein shall be manufactured and tested within the USA or its territories or by a foreign source complying with the provisions of the Defense Standardization Manual, DOD 4120.3-M, Para. 4-106.6, subject to the approval of the US Army ARRADCOM, DRDAR-QAT, Dover, NJ 07801."

3.8 Marking - identification. Each circuit shall be marked with the part number and revision (Example: MIL-A-63242-A2). The circuit shall also be marked with a lot number traceable to a vendor's manufacturing records. The vendor may add additional markings for his use provided the additional markings do not interfere with the required markings.

3.8.1 Inspection lot formation (lot identification). To meet the requirements for material traceability required by contract, each integrated circuit lot will contain only one interfix number.

3.8.2 Resistance to solvent. The marking shall remain legible when subjected to the resistance to solvents testing of MIL-STD-883, Method 2015 except paragraph 2.1.d.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Quality conformance inspection.

4.2.1 Inspection lot formation. In addition to the criteria established in MIL-A-48078, inspection lots of operational amplifiers as described in this detailed specification shall have been manufactured on the same production line(s) through final seal by the same production techniques and to the same device design rules and package with the same material requirements and within the same period not exceeding six (6) weeks.

"The lot identification/date code number shall be preceded by a three-digit interfix number. The interfix number shall be an integral part of the lot identification number, and must be used to identify those lots produced homogenously in one continuous process. Any of the conditions described below shall be cause for changing the interfix number.

Change in manufacturing process. Changes representative of those which are subject to this requirement are any basic design changes in the manufactuing processing such as:

a. Die structure topograph changes (e.g., double-diffused, epitzxial, isolation).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:28Z
Check the source to verify that this is the current version before use.

b. Mask changes or redesign which alter die size.

c. Change in passivation or glassification material thickness or technique (including addition or deletion of passivation).

d. Change in die or substrate or element attach material, method or loclation.

e. Changes in wire dimensions.

f. Package or lid structure and material changes or changes to the internal cavity geometry.

Change in manufacturing location. If the active line moves from one geographical location to another, the interfix number will be changed.

Interruption in manufacturing process over 90 days. A complete line shutdown of more than 90 days duration will be cause for a change of the interfix number.

Change in item design. Only changes involving function and compliance to specifications will be cause for change of the interfix number. Changes made for clarification, to provide additional information, or changes for producibility will not be cause for change of the interfix number.

4.2.2 Sampling. Inspection samples of operational amplifiers required by this detailed specification shall be randomly selected from each inspection lot in accordance with appendix B of MIL-M-38510.

4.2.3 Screening. Screening shall be in accordance with Table II and the MIL-STD-883 procedures specified therein. Screening shall be conducted on all device prior to the final electrical tests.

4.2.4 Electrical tests.

4.2.4.1 Ambient electrical test. Each operational amplifier shall be electrically tested at 25°C ± 3°C for the requirement of Table I. The test shall be performed as specified in 4.3.4.

4.2.4.2 Operating temperature electrical test. A sample of each lot of operational amplifiers shall be subjected to the electrical test reqwuirements of Table I at the operating temperature extremes. An equal quality shall be tested at -40°C and +60°C (±3°C). Test shall be performeld in accordance with 4.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:28Z
Check the source to verify that this is the current version before use.

4.2.5 Bond strength. A sample from each lot of operational amplifiers shall be subjected to the bond strength test. The test shall be performed in accordance with para 4.3.4. At the option of the manufacturer, the test sample may be selected randomly, immediately following internal (pre-cap) visual inspection prior to sealing.

4.2.6 Solderability. A sample from each lot of operational amplifiers shall be subjected to the solderability test. The test shall be performed in accordance with para 4.3.6. This test shall be performed after the components have been subjected to the screening procedures and final electrical test. Units that failed final electrical test may be used.

4.2.7 Die shear strength. A sample from each lot of Operational Amplifiers shall be subjected to the die shear tests as specified in MIL-STD-883, Method 2019.

4.2.8 Physical Dimensions. A sample from each lot shall be pulled and onform to the requirementls of 3.1.1 and be tested per MIL-STD-883, Method 2016.

4.2.9 Marking. A sample from each lot shall be pulled and conform to the requirements of paragraph 3.8. They shall be tested visually and for Resistance to Solvents per MIL-STD-883, Method 2015, except paragraph 2.1.d.

4.2.10 External visual. A sample from each lot lshall be pulled and tested per paragraph 14.3.7.

4.2.11 Thermal shock. Parts shall be Thermal Shocked per MIL-STD-883, Method 1011, Condition C, 15 Cycles.

4.3 Test methods and procedures.

4.3.1 Test method deviation. Deviations from test methods or test circuits specified herein are allowed provided that it is demonstrated that such deviations in no way relax the requirements of this specification and that they are approved before testing is performed. Proposed deviations of test equipment and/or test circuit shall be submitted for approval in accordance with 16.2.

4.3.2 Conditions. Unless otherwise specified, all examinations and tests shall be made under the following conditions:

Temperature - $25 \pm 3^{o}C$

Humidity - 20 to 60 percent RH

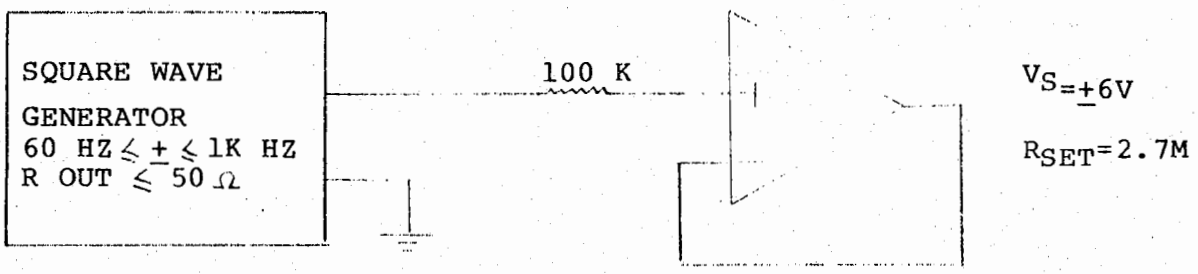
Pressure - 29 ± 2 inches of HG

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:28Z
Check the source to verify that this is the current version before use.

4.3.3 Screening. Screening of the components shall be performed in accordance with the MIL-STD-883 methods as listed in Table II.

4.3.3.1 Burn-in test. The dynamic burn-in shall conorm to the following:

Each operational amplifier shall be connected in the circuit below (fig. 2) and tested per MIL-STD-883, Method 1015, 125°C for 168 hours minimum.



4.3.4 Electrical tests. The test methods for the electrical requirements shown in Table I shall be performed using the test circuit shown in MIL-M-38510/101, fig. 8, or approved equivalent (see 4.3.1).

4.3.5 Bond strength. Bond strength test shall be performed in accordance with MIL-STD-883, method 2011, test condition D.

4.3.6 Solderability. Solderability tests shall be performed in accordance with MIL-STD-883, Method 2003.1.

4.3.7 External visual. External visual shall be performed in accordance with MIL-STD-883, Method 2009.

5.

5.1 Preservation, packaging and packing. The operational amplifiers shall be prepared for delivery in accordance with the preservation, packaging and packing requirements of MIL-M-55565, Level C unless otherwise specified in the procurement document.

6. NOTES

6.1 Ordering date. See MIL-A-48078.

6.2 Submission of designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, Armament Research and Development Command, ATTN: DRDAR-QAA-I, Dover, NJ 07801.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:28Z
Check the source to verify that this is the current version before use.

6.3 Approved sources of supply. Only the item described herein procured from the vendors listed hereon is approved by ARRADCOM for use in mine electronic assemblies. A substitute item shall not be used without prior approval by ARRADCOM and only after successful completion of the qualification testing described in drawing 9317646.

Identification of the approved source herein is not to be construed as a guarantee of present or continued availability as a source of supply for the item described inthis specification.

Approved source of supply

Vendor(s)	Code Ident. No.
Intersil, Inc. Cupertino, CA Ref Part No. ITS 6365	32293
Honeywell, Inc. Solid State Electronics Div. 12001 State Highway 55 Plymouth, MN 55441 Ref Part No. 22002904	
Silicon General 11651 Monarch Street Garden Grove, CA 92641 Ref Part No. SG9804/1253J	

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1345-A215)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:28Z
Check the source to verify that this is the current version before use.